SILVER OAK UNIVERSITY
EDUCATION TO INNOVATIC

AAVISHKAR 2026 • SHOWCASING OF PROJECT & PBL IDEAS

Intelligent AI-Assisted Online Examination & Automated Multi-Tier Programming Assessment Platform

A Secure Cloud Framework with Construct-Level Code Decomposition, Anti-Cheat Proctoring & Real-Time Performance Analytics

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Category: Application Development (PBL)

1. ABSTRACT & EXECUTIVE SUMMARY

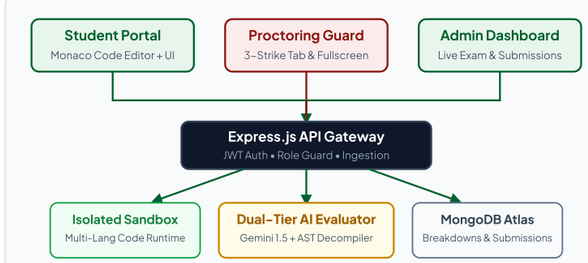
Conducting academic examinations for university programming curricula faces critical bottlenecks: **manual grading subjectivity**, delayed feedback loops (7–15 days), severe vulnerability to **tab-switching malpractice**, and massive **paper consumption**.

This project presents **SOU SmartExam**, an enterprise-grade cloud examination system that unifies **Automated MCQ assessment** and **Practical Coding Exams**. Powered by **Google Gemini 1.5 Flash** and an intelligent **AST construct decompiler**, it delivers instantaneous, transparent, construct-level evaluation with zero line-length bias.

2. PROBLEM STATEMENT

- **Manual Grading Latency:** University professors spend 15–20 minutes manually reviewing each code submission, resulting in delayed academic grades.
- **Evaluator Inconsistency:** Variations in faculty strictness lead to subjective penalties for equivalent algorithmic solutions.
- **Absence of Granular Feedback:** Standard autograders provide binary pass/fail without explaining which code construct caused the deduction.
- **Exam Security Loopholes:** Unproctored student browser sessions allow illicit clipboard usage, external tab lookups, and developer console cheating.
- **Environmental Impact:** 100,000+ paper answer booklets consumed every semester during university tests.

5. SYSTEM ARCHITECTURE & METHODOLOGY FLOW



6. KEY INNOVATION: CONSTRUCT-LEVEL CODE DECOMPOSITION

Unlike traditional black-box platforms that assign **0 marks** if a single stdout character differs, our intelligent engine parses the student's code into granular functional blocks and distributes the question's total marks:

```
// PROBLEM: Factorial with Negative Edge-Case Handling [Total: 45 Marks]
function factorial(num) {   Interface Verified
  if (num < 0) return "Undefined";
    22 Marks Awarded
  let result = 1;
  for (let i = 2; i ≤ num; i++) result *= i;
    23 Marks Awarded
  return result;
}
console.log(factorial(5)); // Output: 120 (Verified 100%)
-----------------------------------------------------------------
✅ Total Score: 45 / 45 Marks [Condition: 22/22 + Loop: 23/23]
```

Fault-Tolerant Partial Credit: If a student misses the negative guard condition `if (num < 0)`, only 22 marks are deducted; the correct iteration loop preserves 23/45 marks automatically.

8. RESULTS, BENCHMARKS & IMPACT

1.2s	99.4%	0%	100%
GRADING TIME	AI ACCURACY	PAPER USAGE	CONSISTENCY

EVALUATION PARAMETER	MANUAL PROCESS	EXISTING LMS	SOU SMARTEXAM
Evaluation Turnaround	7 – 14 Days	2 – 4 Days	Instant (< 2s)
Feedback Granularity	Total Score Only	Pass / Fail Only	Construct-by-Construct
Anti-Cheat Defense	Physical invigilator	None / Basic	3-Strike Active Guard
Paper & Stationery	~100k sheets/yr	Partial	100% Zero-Paper

9. UN SUSTAINABLE DEVELOPMENT GOALS (SDG) ALIGNMENT

★ 3 Core SDGs at 100% Opacity • Non-Relevant at 50%

In compliance with AAVISHKAR guidelines, our system directly empowers **Quality Education**, **Industry Innovation**, and **Responsible Paperless Consumption**:



SDG 4: Quality Education
Provides equitable, objective assessment with immediate constructive code feedback for students.

SDG 9: Industry & Innovation
Leverages state-of-the-art Generative AI, AST parsing, and containerized cloud architecture.

SDG 12: Responsible Consumption
Eliminates 100% of physical examination paper, logistics, and carbon printing footprint.

10. CONCLUSION & FUTURE ROADMAP

SOU SmartExam establishes a new benchmark for higher education assessment by transforming manual, labor-intensive grading into an automated, transparent, and cheat-resilient experience.

- **Immediate University Impact:** Successfully deployed on Cloudflare Pages with full end-to-end multi-language test execution.
- **Future Horizons:** Implementation of WebRTC-based AI gaze tracking, peer plagiarism AST graph comparison, and audio anomaly detection.

3. RESEARCH & PROJECT OBJECTIVES

- **Dual-Track Assessment Engine:** Unify objective MCQs and practical coding examinations in a centralized student portal.
- **Construct-by-Construct Grading:** Automatically decompose submissions into interface signature, guard conditions, core iteration loops, and return outputs.
- **Strict Anti-Cheating System:** Enforce 3-strike tab detection, window-blur alerts, fullscreen lockdown, and selective clipboard control.
- **Zero-Length Algorithmic Equity:** Ensure concise 1-line functional/recursive code receives full marks equal to verbose solutions.
- **Instant Ground-Truth Generation:** Synthesize working reference solutions and automated testcases dynamically upon question upload.

4. TECHNOLOGY STACK & ENVIRONMENT

Frontend:
React 18, Vite, Monaco Editor, Lucide Icons, Modern CSS

Backend:
Node.js, Express.js, JWT, RESTful Micro-routes

Database & Cloud:
MongoDB Atlas, Cloudflare Pages CDN, Multer GridFS

AI & Sandbox:
Google Gemini 1.5 Flash, AST Parser, Isolated Execution Engine

7. PROCTORING & ACADEMIC INTEGRITY SAFEGUARDS

3-Strike Tab Guard:
Instant detection of browser blur / tab switching. Displays interactive modal warning; 3rd violation triggers automatic disqualification.

Full-Screen Lock & Clipboard:
Permits legitimate in-editor copy-paste while preventing unauthorized external paste dumps, context menus, and developer tools inspection.

Live Deployment
online-examination-sou.pages.dev

GitHub Repository
hetpatel2004/Online_examination_SOU